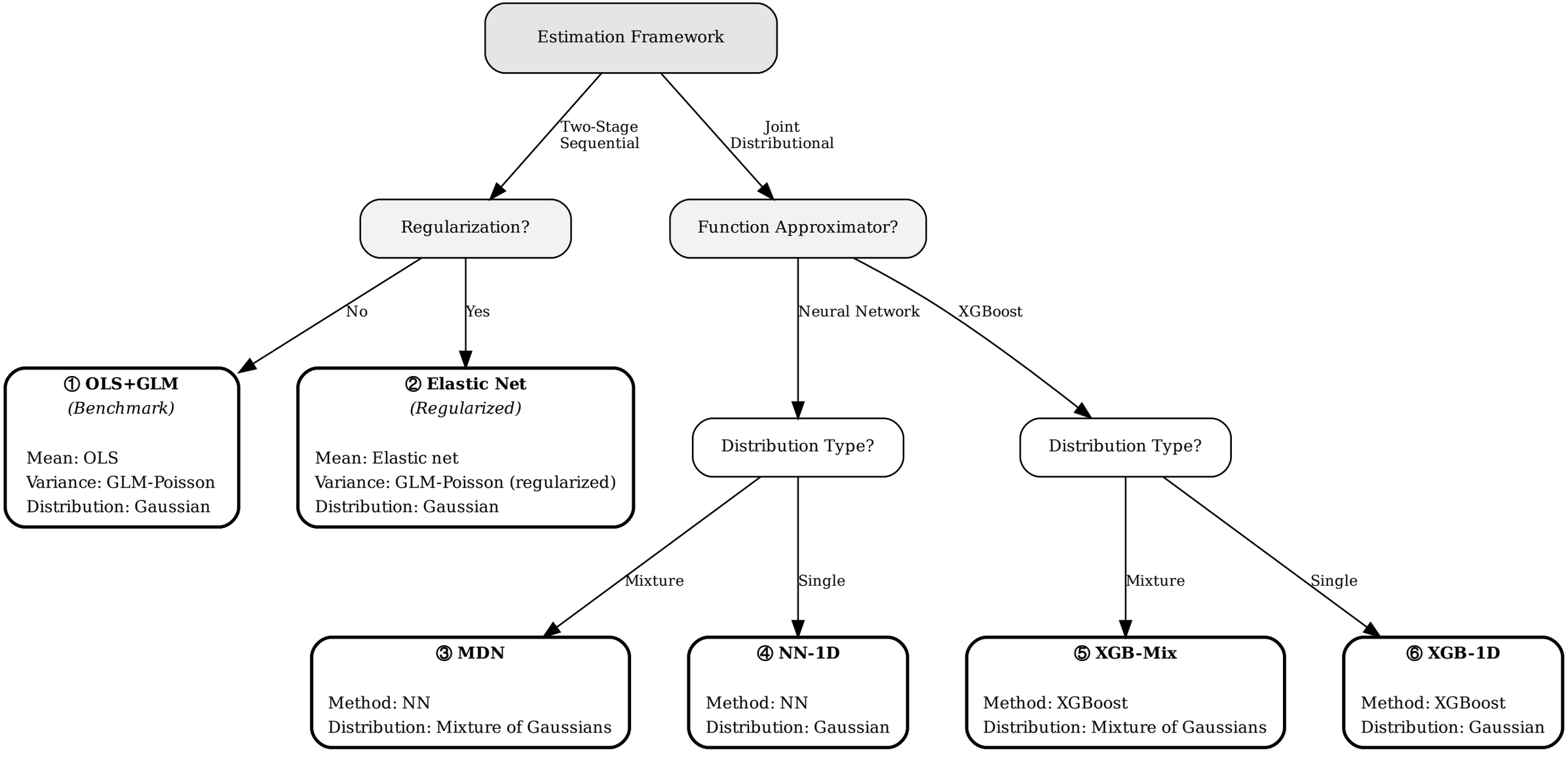
Estimation Framework
Two-Stage
Sequential
Joint
Distributional
Regularization?
Function Approximator?
No
Yes
Neural Network
XGBoost
① OLS+GLM
(Benchmark)
Mean: OLS
Variance: GLM-Poisson
Distribution: Gaussian
② Elastic Net
(Regularized)
Mean: Elastic net
Variance: GLM-Poisson (regularized)
Distribution: Gaussian
Distribution Type?
Distribution Type?
Mixture
Single
Mixture
Single
③ MDN
Method: NN
Distribution: Mixture of Gaussians
④ NN-1D
Method: NN
Distribution: Gaussian
⑤ XGB-Mix
Method: XGBoost
Distribution: Mixture of Gaussians
⑥ XGB-1D
Method: XGBoost
Distribution: Gaussian